Yecheng WANG

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EDUCATION

University of Leeds, Leeds, UK — Sep 2023 – Jul 2027
BEng (Hons) Mechanical Engineering, expected Jul 2027
- **GPA 3.92/4.0**, average 78.67/100 (on track for First Class)
- Final-year project: learning-based control for rehabilitation robots, Rehab45 Lab (supervisor: Dr. Justin Gallagher)

Southwest Jiaotong University (SWJTU), Chengdu, China — Sep 2023 – Jul 2027
*BEng Mechanical Design, Manufacturing and Automation, SWJTU-Leeds Joint School (**2+2**: Chengdu 2023–25, Leeds 2025–27)*
- **GPA 3.89/4.0**, average 92.7/100
- **National Scholarship**, Ministry of Education of China (top 0.2% nationally, 2 per cohort/year), **awarded twice**: 2024 & 2025

PUBLICATIONS

1. **Y. Wang**, Y. Li, "PHLEX: Physically Layered Explanation for Counterfactual Motion Planning," ***ICLR 2027*, under review**.
2. **Y. Wang**, Y. Li, "KineWorld: A Pluggable Counterfactual Kinematic World Model for Physically Grounding Multimodal Motion Prediction," ***under review***.
3. **Y. Wang**, J. Huo, G. Zhang, S. S. Cheng, "Task-Space Capability of 5-DoF Continuum Instruments from Surgical Gestures," ***in preparation***.
4. **Y. Wang**, "MBO-YOLO: An Enhanced YOLOv8n-Based Algorithm for Detecting Marine Benthic Organisms," in *AIITA 2025*, Xi'an, China, pp. 854–860. DOI: 10.1109/AIITA65135.2025.11047694.

RESEARCH EXPERIENCE

Surgical Robotics & Instrumentation Lab (SRIL), CUHK, Hong Kong SAR — Jun 2026 – Present
Research Intern · PI: Prof. Shing Shin CHENG · Direct Mentor: Dr. Jun HUO

P1: From Millimetres to Microns: Hysteresis Compensation of a Cable-Driven Continuum Surgical Robot
CUHK Summer Undergraduate Research Programme · Poster: yecheng-wang.com/cuhk-surp-2026 — Jun 2026 – Aug 2026
- Rebuilt a 14-notch tendon-driven continuum surgical robot with stereo tip tracking and cable-tension sensing and identified path-dependent cable hysteresis as the dominant error source.
- Designed a data-driven hysteresis compensator and a P-type iterative learning controller for the 0.25 s vision latency; showed that tip loading widens the hysteresis loop, contradicting a friction-only assumption.
- Validated the frozen compensator 18 days later on a 28-target test sequence, cutting tip-error scatter 3.0× (234 → 78 µm RMS); three ILC iterations on that sequence reached 15 µm, near same-day measurement noise.

P2: Task-Space Capability of 5-DoF Continuum Instruments from Surgical Gestures — Remote · Aug 2026 – Present
- Built a pipeline that tests thousands of recorded dVRK gestures on 4- and 5-DoF continuum instruments under position, tool-axis, joint-range and rate limits, adding a verified path search that exposed the sequential solver's missed solutions.
- Attributed each failure to reach, tool-axis or rate limits and located the joint limits involved; wrote the study's technical report and proposed open-loop trajectory replay to test the per-gesture predictions on hardware.

Physically Grounded Counterfactual World Models for Explainable Motion Prediction & Planning — Dec 2025 – Present
Independent research · with a Ph.D. student at Purdue University

KineWorld: A Pluggable Counterfactual Kinematic World Model for Physically Grounding Multimodal Motion Prediction
- Proposed a pluggable world model that decodes a frozen predictor's trajectories against a library of differentiable motion mechanisms, giving interpretable behavior attribution and physically grounded counterfactual rollouts.
- Grounded a frozen UniAD stack on nuScenes, making its plans more physically feasible and interpretable at no accuracy cost: maximum yaw rate 0.986 → 0.640 rad/s, mechanism self-consistency 8.45% → 57.27%, fewer collisions.

PHLEX: Physically Layered Explanation for Counterfactual Motion Planning
- Extended the library into a predictor-agnostic framework that explains predicted trajectories through layered kinematic models, control laws and scene references, and re-runs varied explanations as counterfactual futures for ego planning.
- Cut collision rate by up to half and yaw-rate violations by 1.9–6.7 pp on three frozen planners (UniAD, VAD, SparseDrive).
- Reduced kinematic violations in agent forecasting (UniAD/MTR/QCNet, nuScenes/Argoverse 2) by >80% for ≤0.05 m minFDE.

MBO-YOLO: Enhanced YOLOv8n for Marine Benthic Organism Detection, Chengdu, China — Oct 2024 – Mar 2025
Independent research
- Integrated a P2 detection head, Non-Local Block attention and BiFPN into YOLOv8n for small, blurred underwater targets: +2.2 pp mAP50 and +2.9 pp mAP50:95 over the baseline on URPC2020. Code: github.com/Geoffrey-WANG-1027/MBO-YOLO.

SELECTED PROJECTS

Magnetic Capsule Robotic Arm Navigation System | MATLAB, C++, Leeds, UK Feb 2026 – May 2026
*MECH2636 Team Project · Kinematics & Trajectory Software Lead · **3rd Place** in the cohort*
- Wrote the inverse-kinematics solver and coordinate-transformation framework for a 2-DoF planar arm steering a magnetic capsule in a colonoscopy simulator, with elbow-down configuration logic and workspace constraints.
- Built the reachability simulation and the pipeline discretizing trajectories into motor counts for embedded C++.

Autonomous Delivery Buggy Control System | C++, Arduino Mega, Leeds, UK Nov 2025 – Feb 2026
*MECH2636 Team Project · Control & Embedded Software Lead · **Champion** of the cohort*
- Designed the embedded control architecture: finite-state mission manager, closed-loop PI velocity control with wheel synchronization and encoder filtering, and on-the-move payload release.
- Tuned gains and anti-windup on ReLOAD and in field trials; moved encoders to non-driven wheels to remove slip-induced velocity error. The competition run used the simpler open-loop FSM configuration for repeatability.
- Contributed the quarter-car (2-DoF mass–spring–damper) stiffness and damping estimates for the team's suspension choice.

Vibration Isolation Platform Design | MATLAB, Leeds, UK Feb 2026 – Mar 2026
MECH2620 Team Project · Team Leader
- Derived a base-excited SDOF isolator's displacement/acceleration frequency responses over 0–12.3 Hz; showed the baseline breached the 9 m/s² limit, then grid-searched stiffness and damping to cut peak acceleration 11.4 → 5.3 m/s².
- Designed a 2-DoF tuned mass absorber (4.03 kN/m at 6.15 Hz), solved the M–K eigenproblem for its natural frequencies and harmonic responses, and recommended the isolator for robustness against the absorber's undamped resonances.

Helicopter Altitude Control | C++, Arduino, Chengdu, China Mar 2025 – Apr 2025
XJME1010 Individual Project
- Implemented and tuned a PID altitude controller for a tethered single-axis helicopter rig, run remotely on the University of Leeds ReLOAD platform: take-off, timed hover within ±2 cm of the target height, and landing, with LED feedback on height error.

YOLO-Based Automatic-Checkout Cart | SolidWorks, Python, PyTorch, Chengdu, China Apr 2024 – Apr 2025
Student Research Training Program (SRTP) · Hardware and Mechatronics Engineer
- Built and deployed a Raspberry Pi 5 cart prototype fusing a fisheye camera and a load cell for weight–vision consistency checks, with a 2-DoF pan-tilt laser pointer that physically flags unrecognized items.
- Integrated a Multi-Scale Dilated Attention module into YOLOv8n: 92.4% mAP@0.5, 91.5% recall, <1 s/item.

INTERNSHIP

KNEWBOTS (Hangzhou Xiaowu Intelligence), Hangzhou, China Jul 2025 – Aug 2025
Mechatronics Engineer Intern
- Calibrated multi-sensor SLAM (LiDAR, ToF camera, ultrasound) on lift and forklift AMRs on the production line; validated mapping and localization by monitoring LiDAR point-cloud distributions in RViz.
- Diagnosed perception failures from mechanical misalignment; regression-tested the ROS-based navigation stack against more than 600 validation metrics and traced deployment edge cases in sensor fusion and loop closure.

SKILLS

Programming & Software: Python, C++, MATLAB/Simulink, LaTeX, Git, OpenCV, ROS
Robot Control & Modeling: PID, ILC, system identification, continuum kinematics, state-machine control
Learning & Perception: PyTorch, YOLO, UniAD, nuScenes
Hardware & CAD/CAE: SolidWorks (CAD), Abaqus (FEA), ANSYS Fluent (CFD), 3D printing, sensor integration and calibration
Language: Mandarin (native), English (IELTS Academic 7.5)

SELECTED MODULES

Completed (Leeds marks, First Class ≥ 70): Engineering Mathematics (91), Computers in Engineering Analysis (91), Vibration and Control (85), Design and Manufacture 1 & 2 (79, 82), Solid Mechanics (81)
In progress (2026/27): Robotics and Machine Intelligence, Computational Fluid and Structural Mechanics, Additive Manufacturing

LEADERSHIP

SWJTU Engineering Practice & Innovation Association, Chengdu, China Nov 2023 – Jun 2025
Co-founder · University Student Organization; 100+ members/year
- Ran the shared workshop (Creative Lab, 3D printing, laser engraving): equipment access, safety and hands-on support.
- Built a recurring project pipeline that created sustained opportunities for students to develop projects and participate in engineering competitions.